

Hands-on Activity 5.4

Gameplay Mechanics	
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Course Code: CPE 411

Program: Computer Engineering

Course Title: Interactive Extended Reality

Date Performed: 16/04/2026

Section: 22S2

Date Submitted: 16/04/2026

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Instructor: Engr. Jimlord M. Quejado

1. Discussion

Finally, push your project further with **Gameplay Mechanics**, where you'll build deeper systems that keep players engaged. You'll implement features such as power-ups with timed buffs, enemy AI with simple state machines, and dynamic difficulty that reacts to player performance. You'll also learn to refactor code into reusable components and scriptable objects, preparing you for larger projects ahead. By completing this activity, you'll transform a basic prototype into a robust, replayable experience ready for user testing and iteration.

2. Materials and Equipment

Unity Hub and Unity Editor

Learn Unity Website ([Unity Essentials Pathway - Learn Game Development for Beginners | Unity Learn](#))

Computer with Network Connection

3. Procedure

✓

1. Unit 4 - Introduction

Q&A (0)

Powerups

Increasing Difficulty

increasing one and then increasing the other in response.

11:00

1x

Mark Step As Incomplete

[Watch the introduction](#)

Unity organization

RossUsesUnity

Project name *

Prototype 4

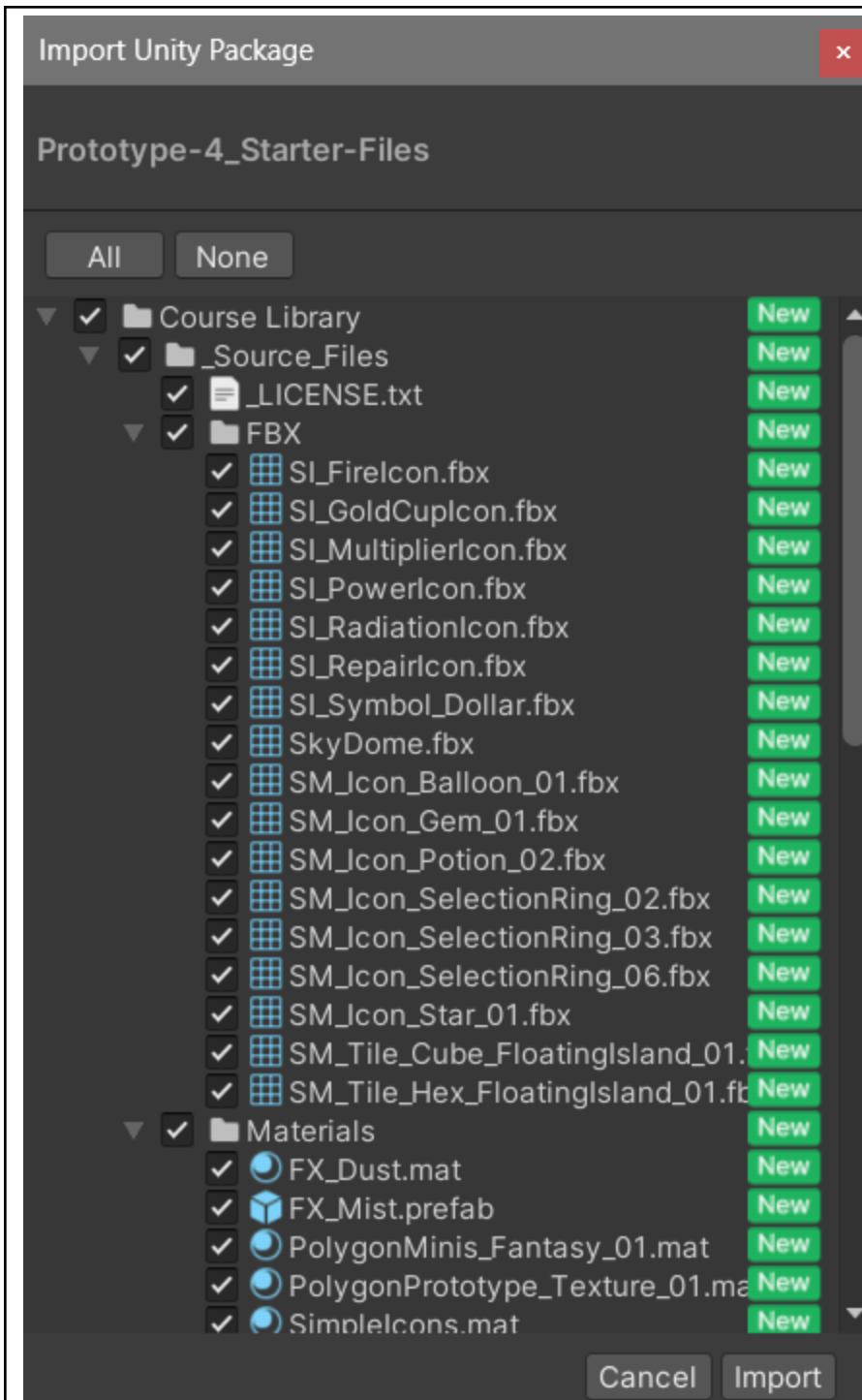
Location *

C:\Users\Andrew Bulaong\Create With ...

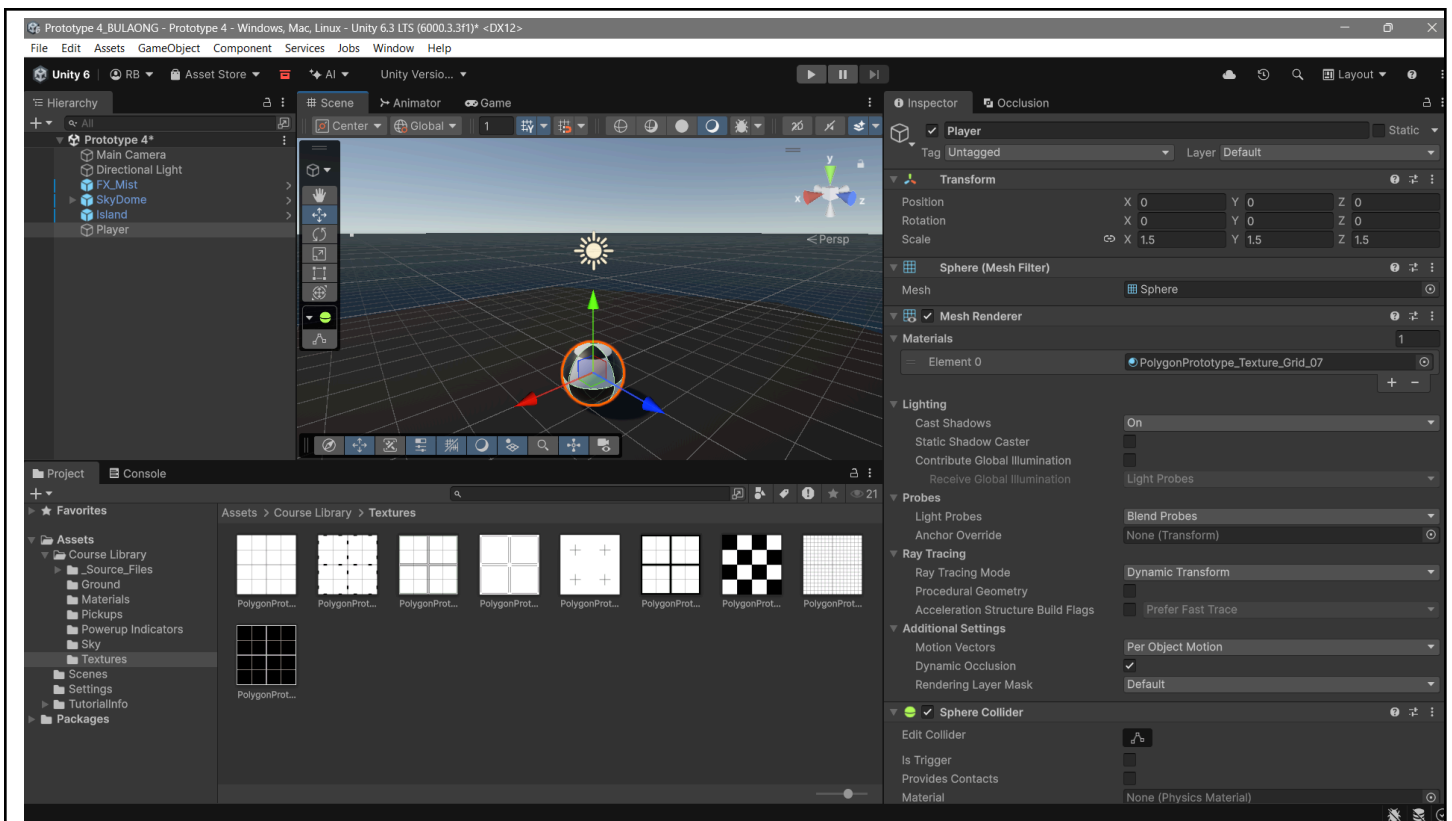
☐ Use Unity Version Control ?

+ Create project

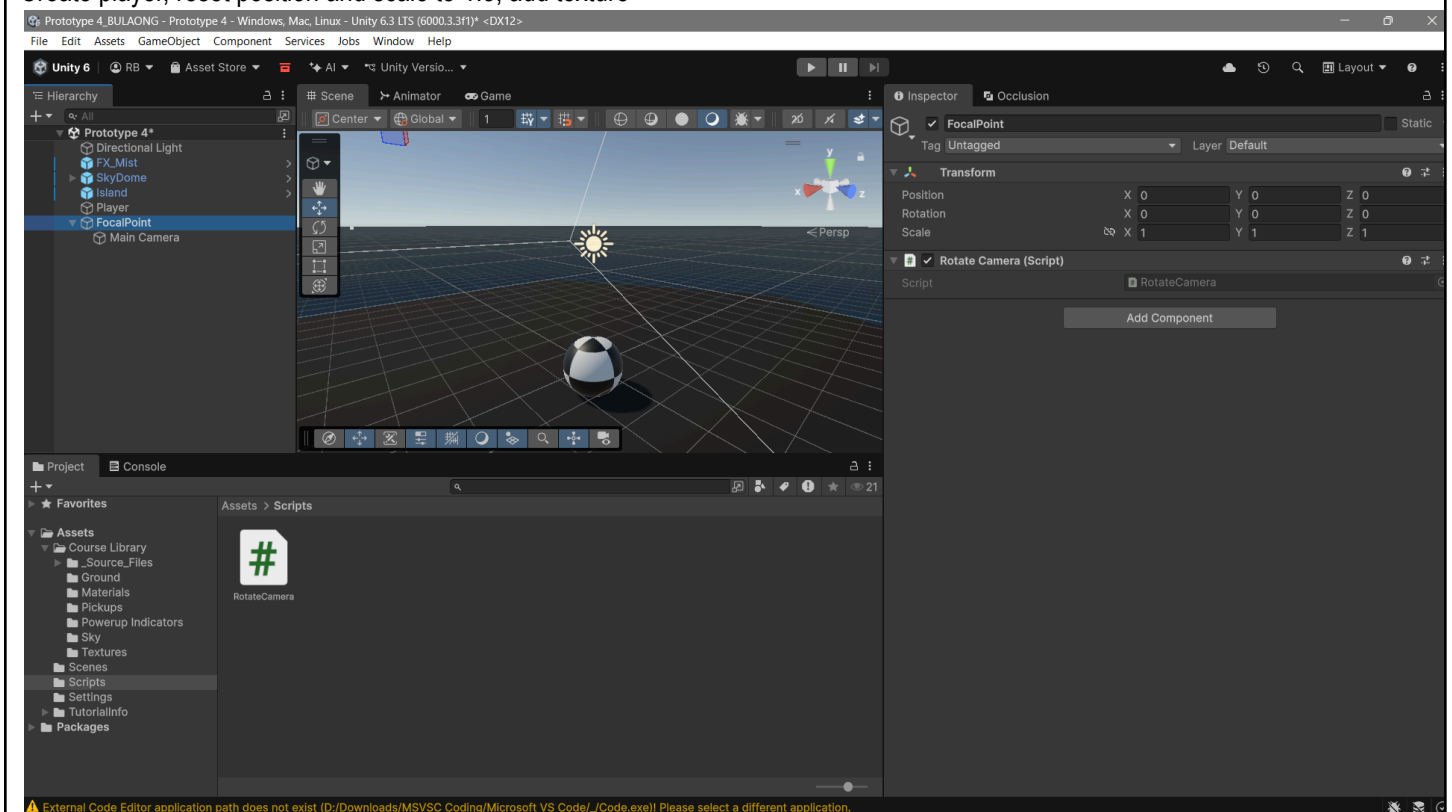
Create project



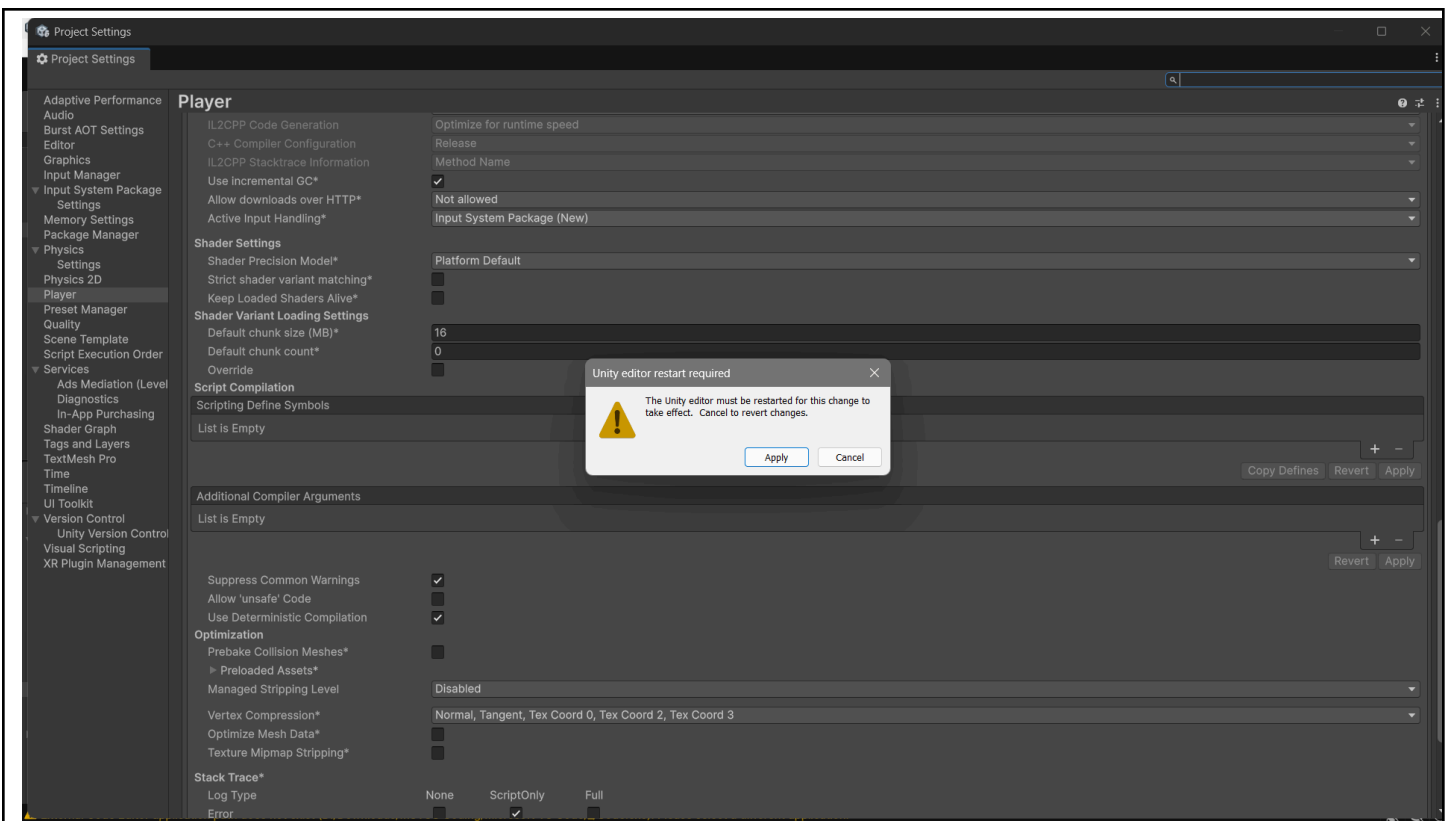
import assets



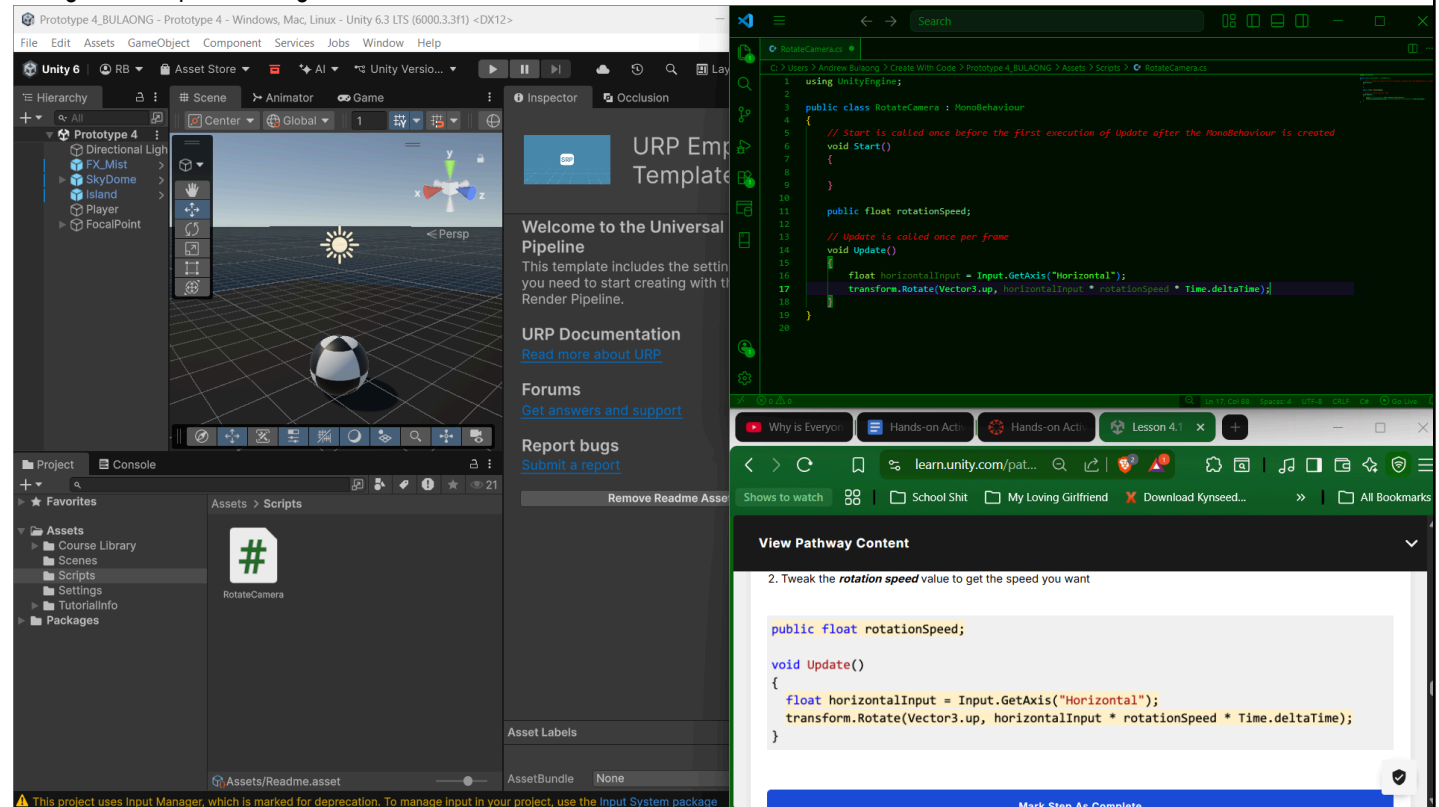
Create player, reset position and scale to 1.5, add texture



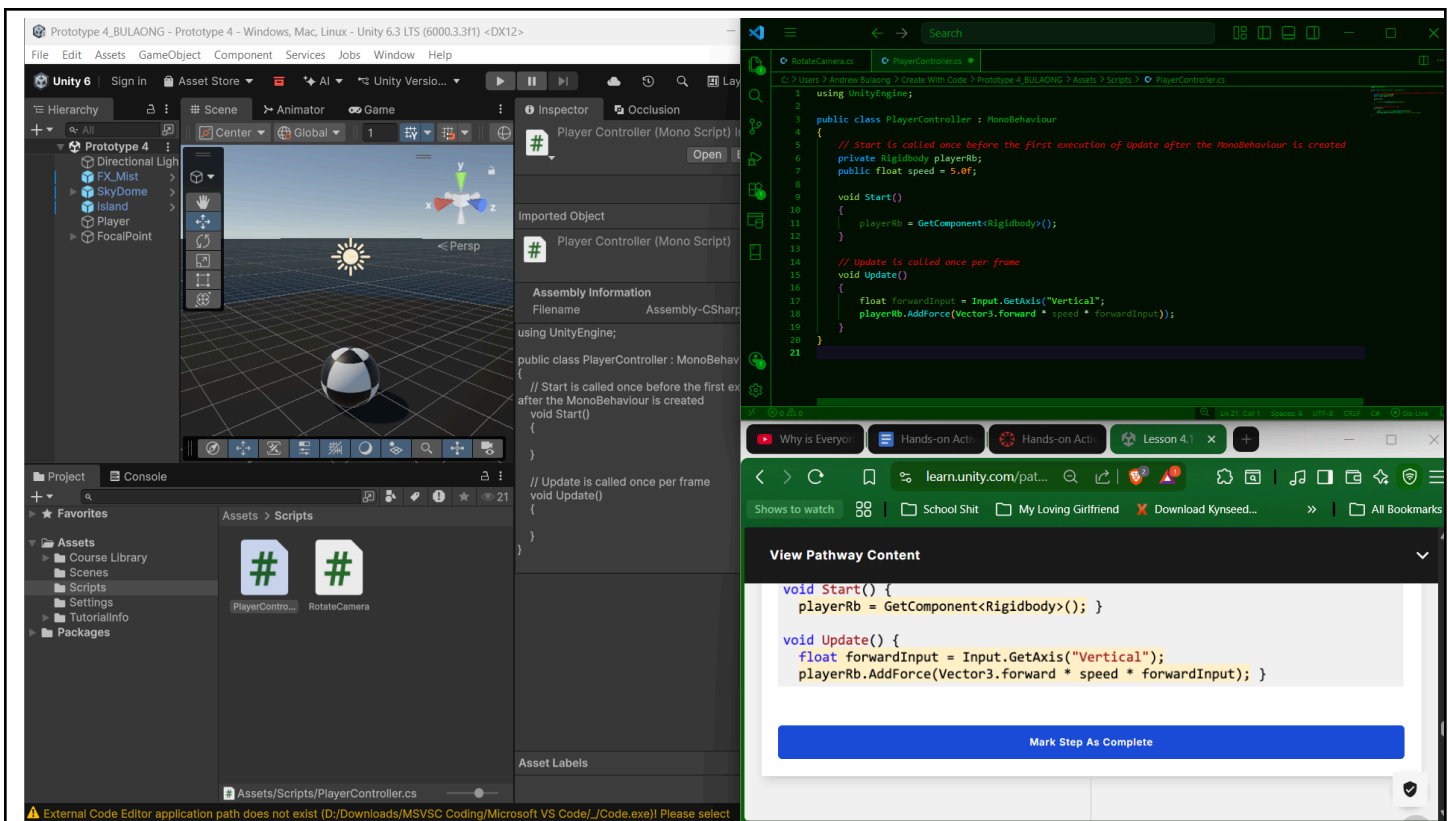
Create focal point game object, attach rotate camera script to it



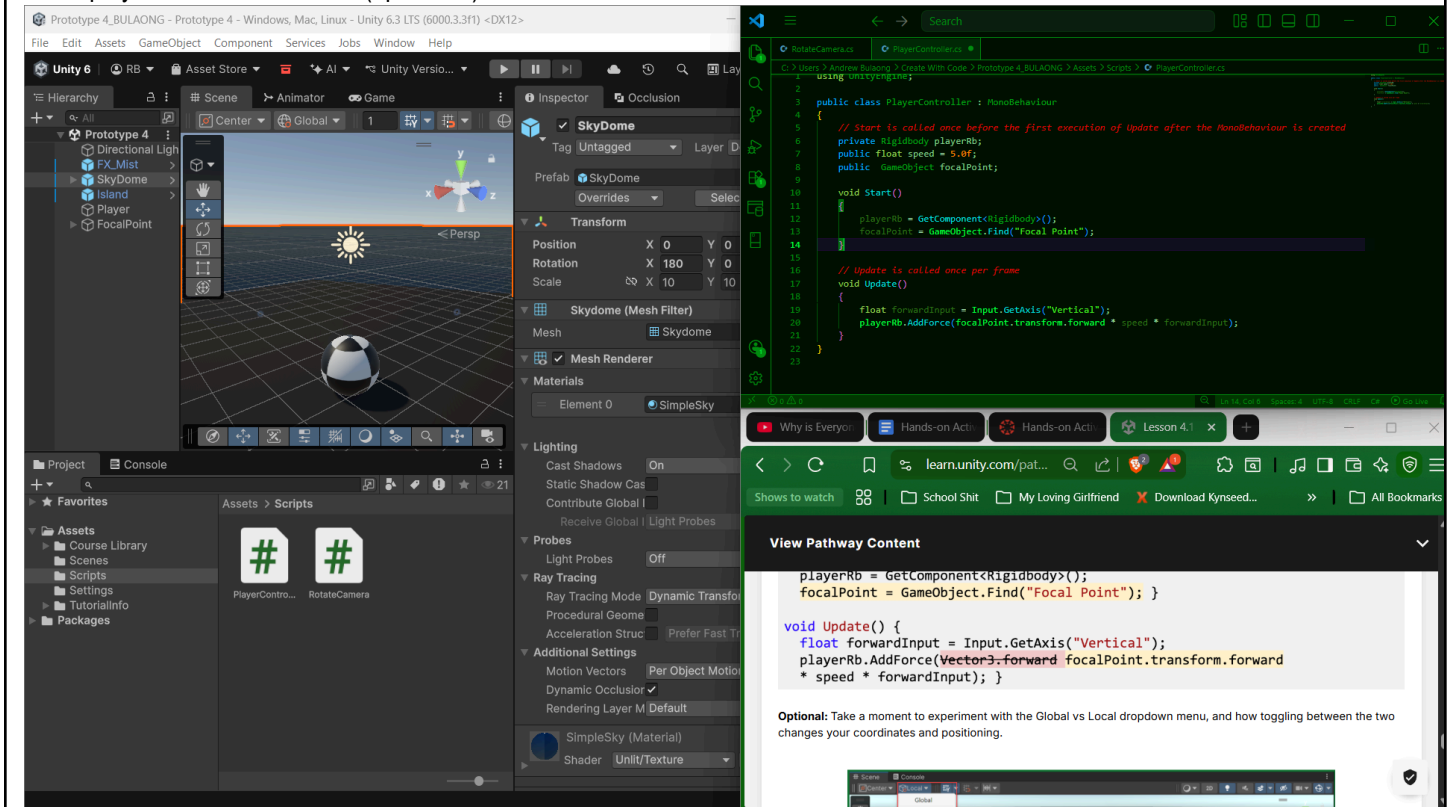
Configure the input handling to "Both"



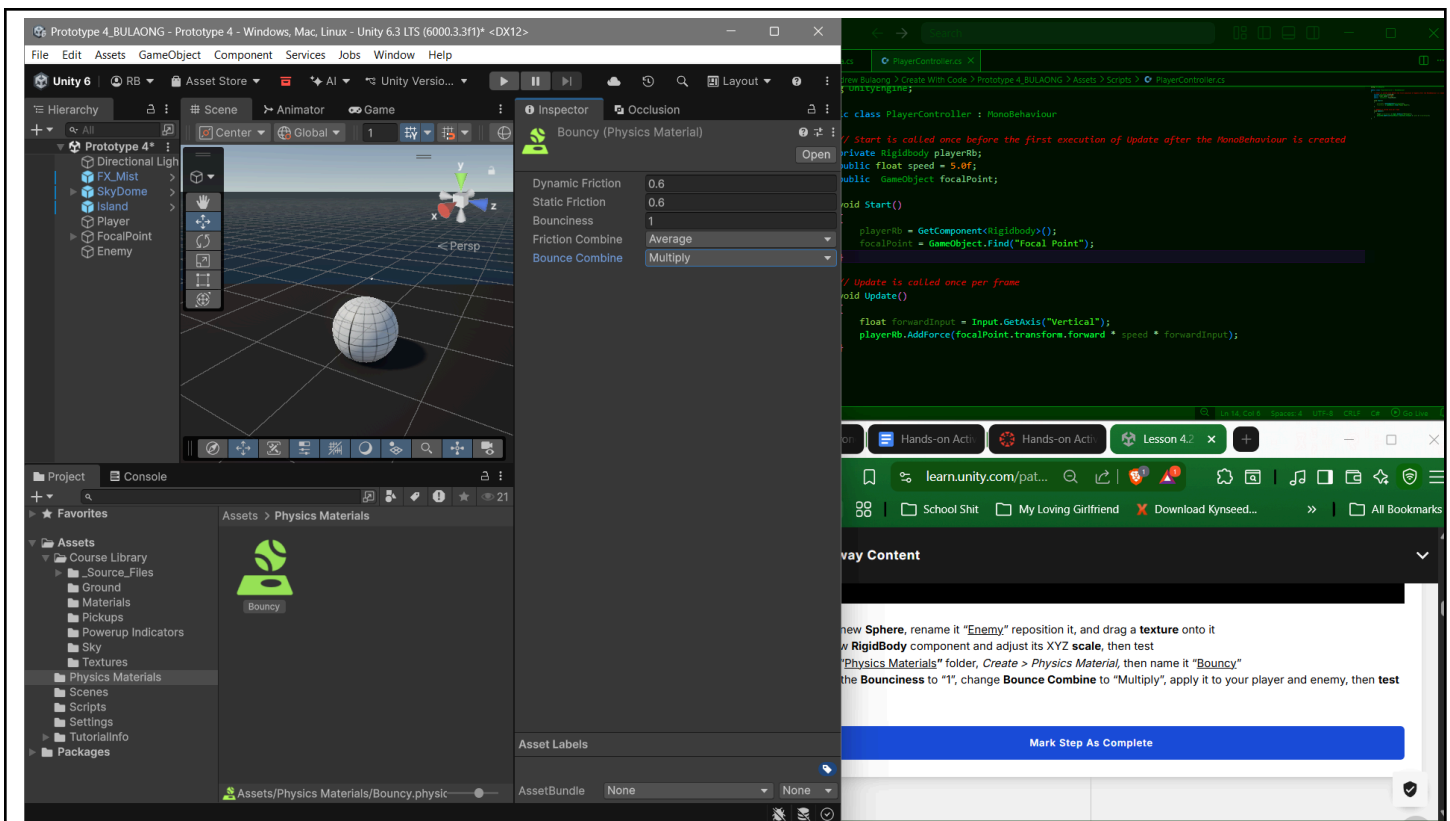
Add the ability to rotate the camera based on rotationSpeed



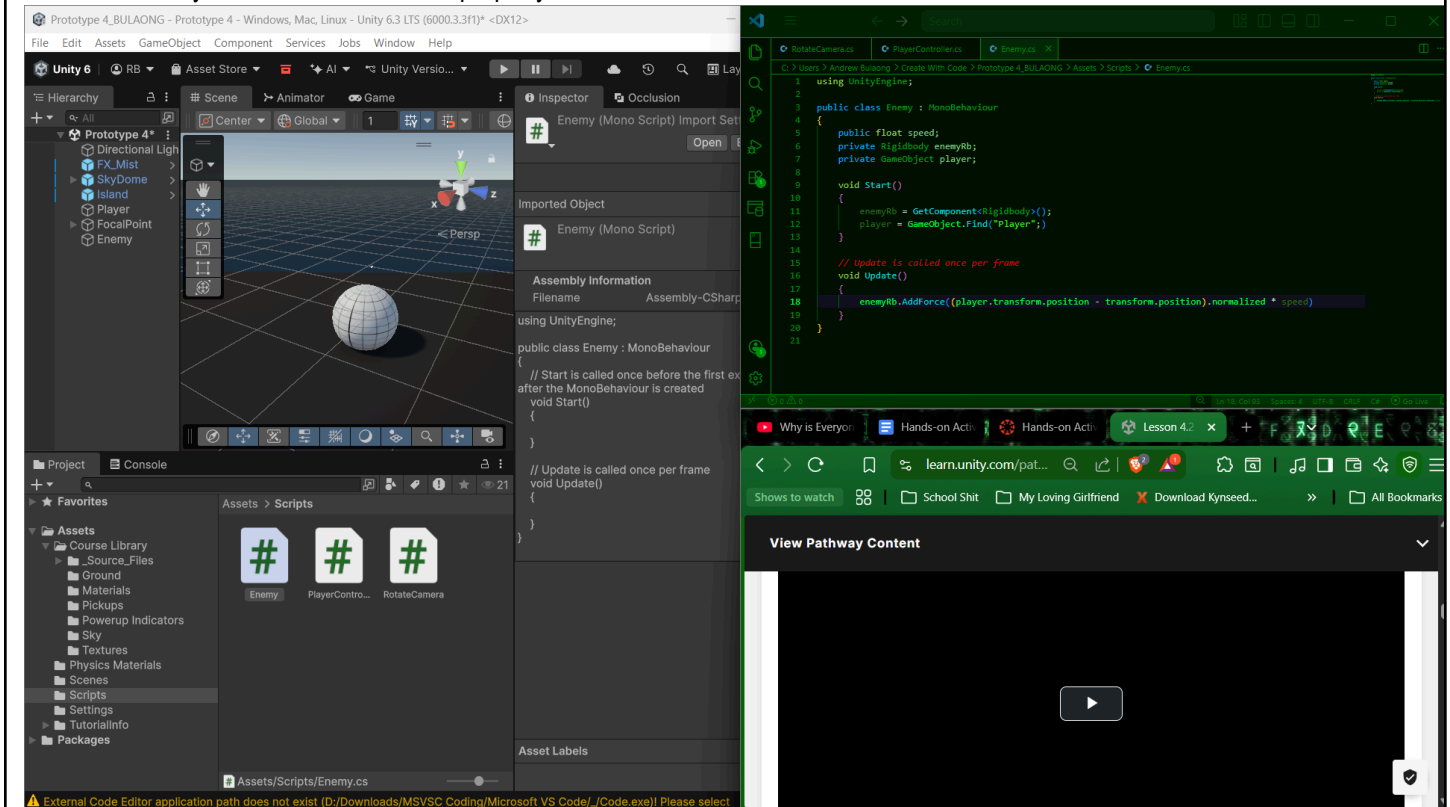
Add a playercontroller + vertical (up button) controller/movement



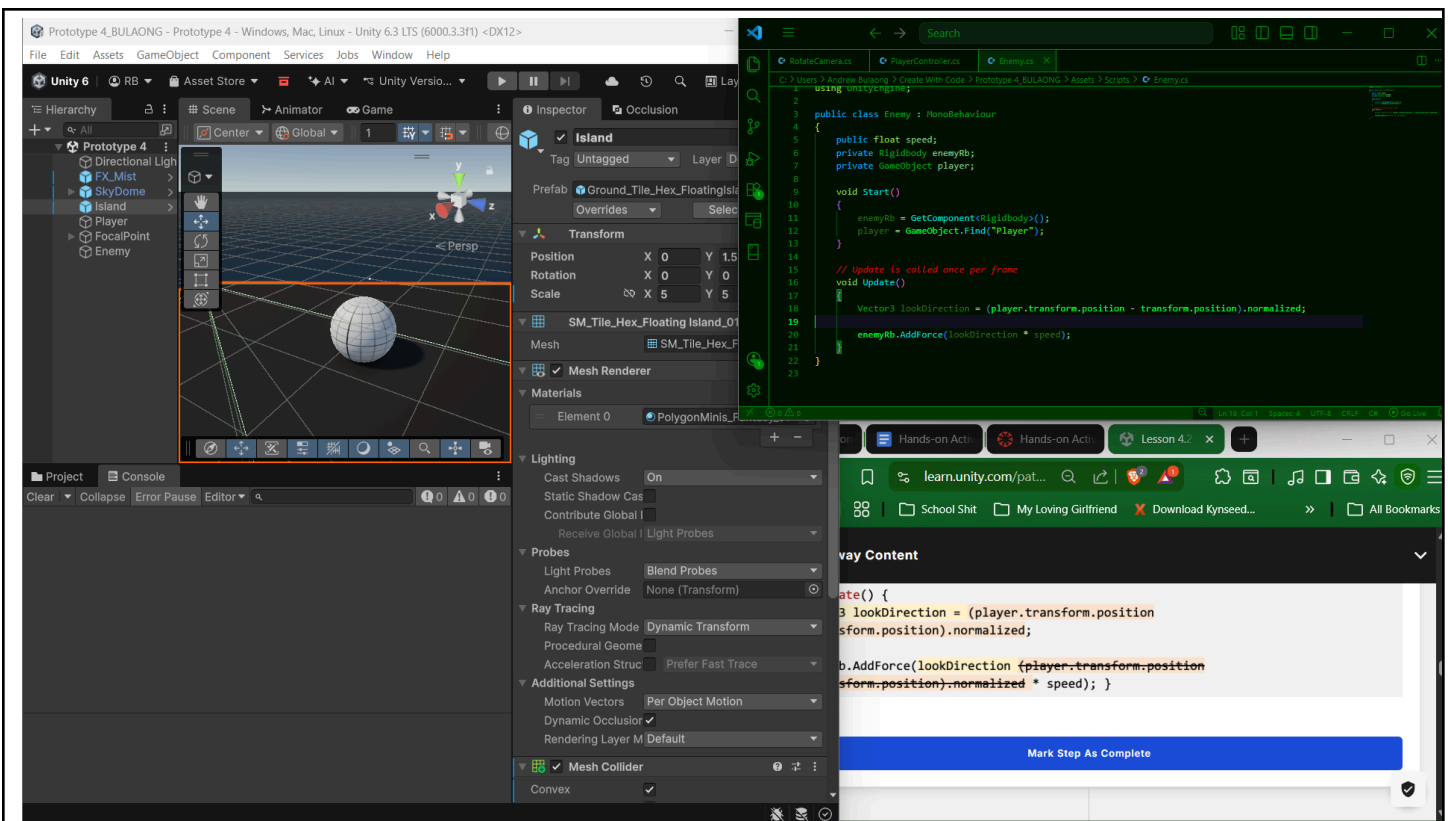
Allow the "Drift" or movement for other directions by using the focal point as the basis



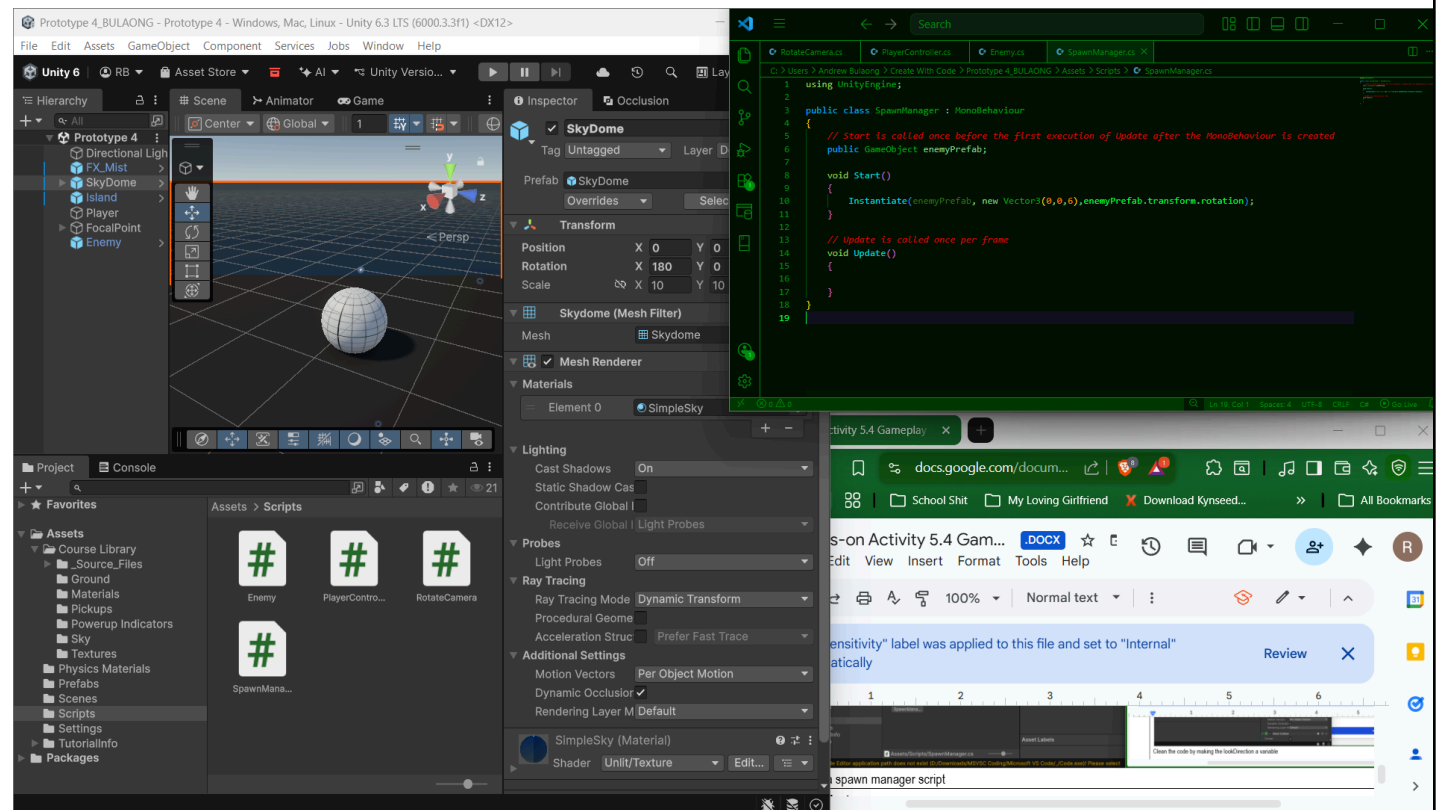
Create an enemy then add a bounciness property to it



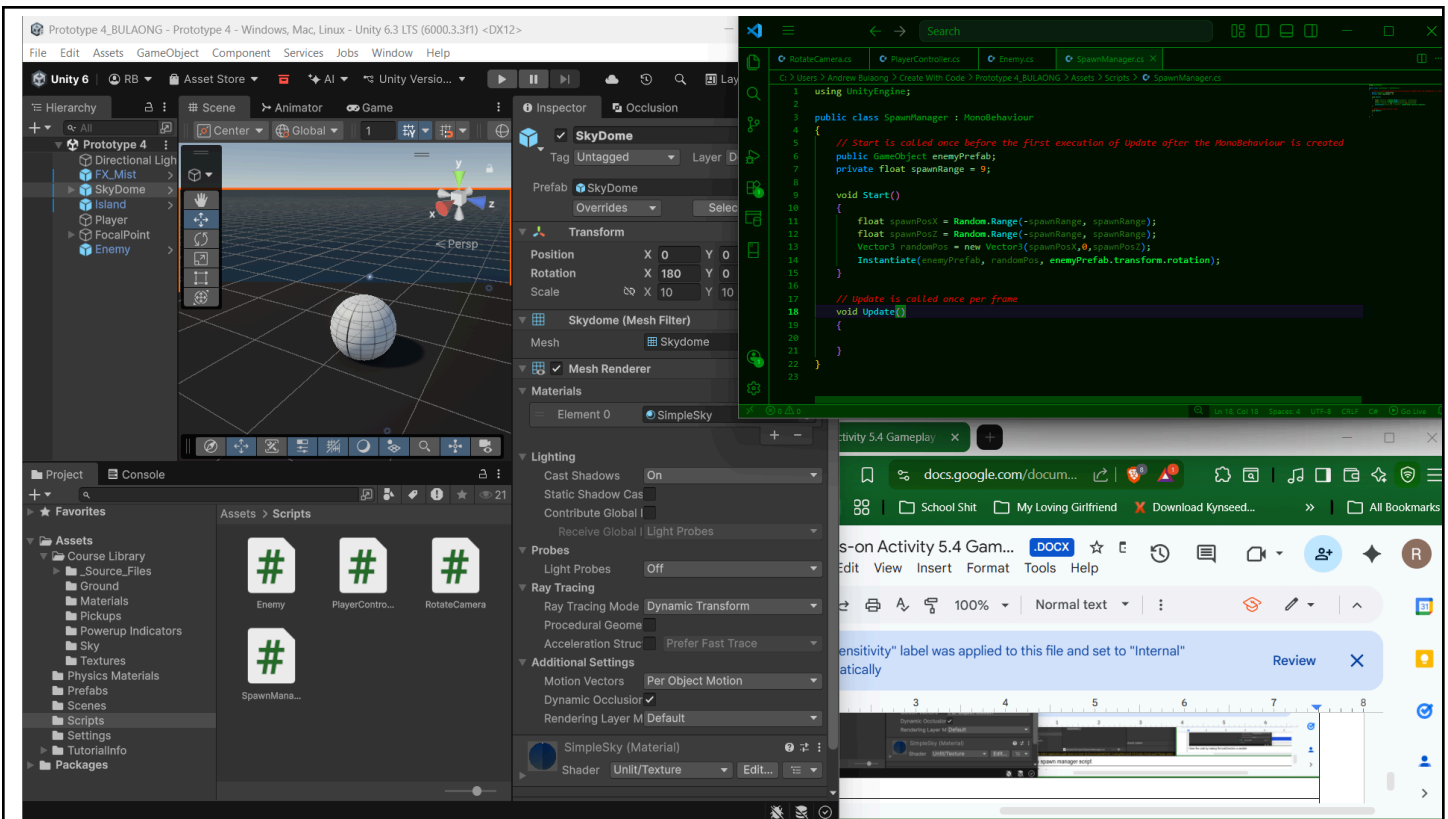
Add a script, that when attached to the enemy, follows the player around the field



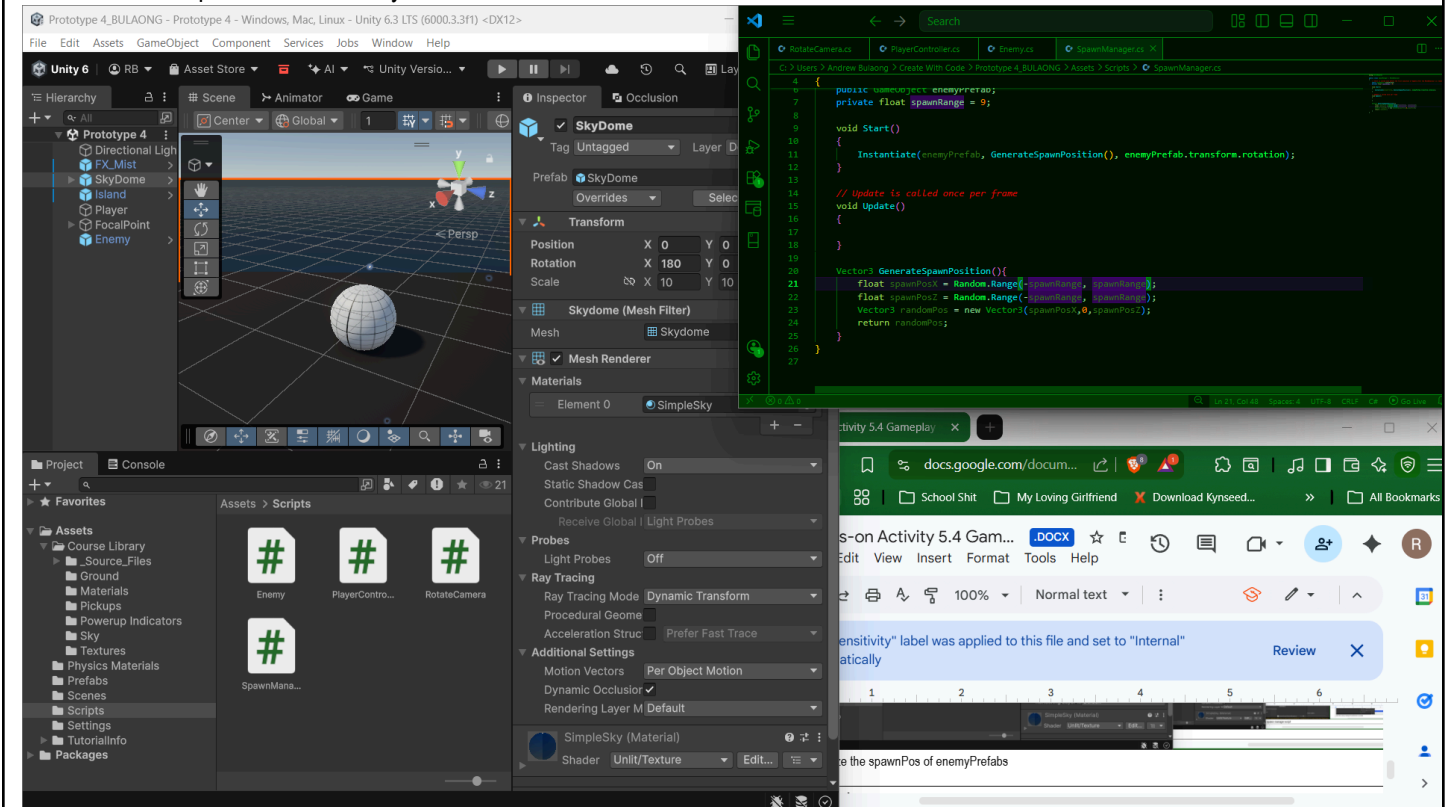
Clean the code by making the lookDirection a variable



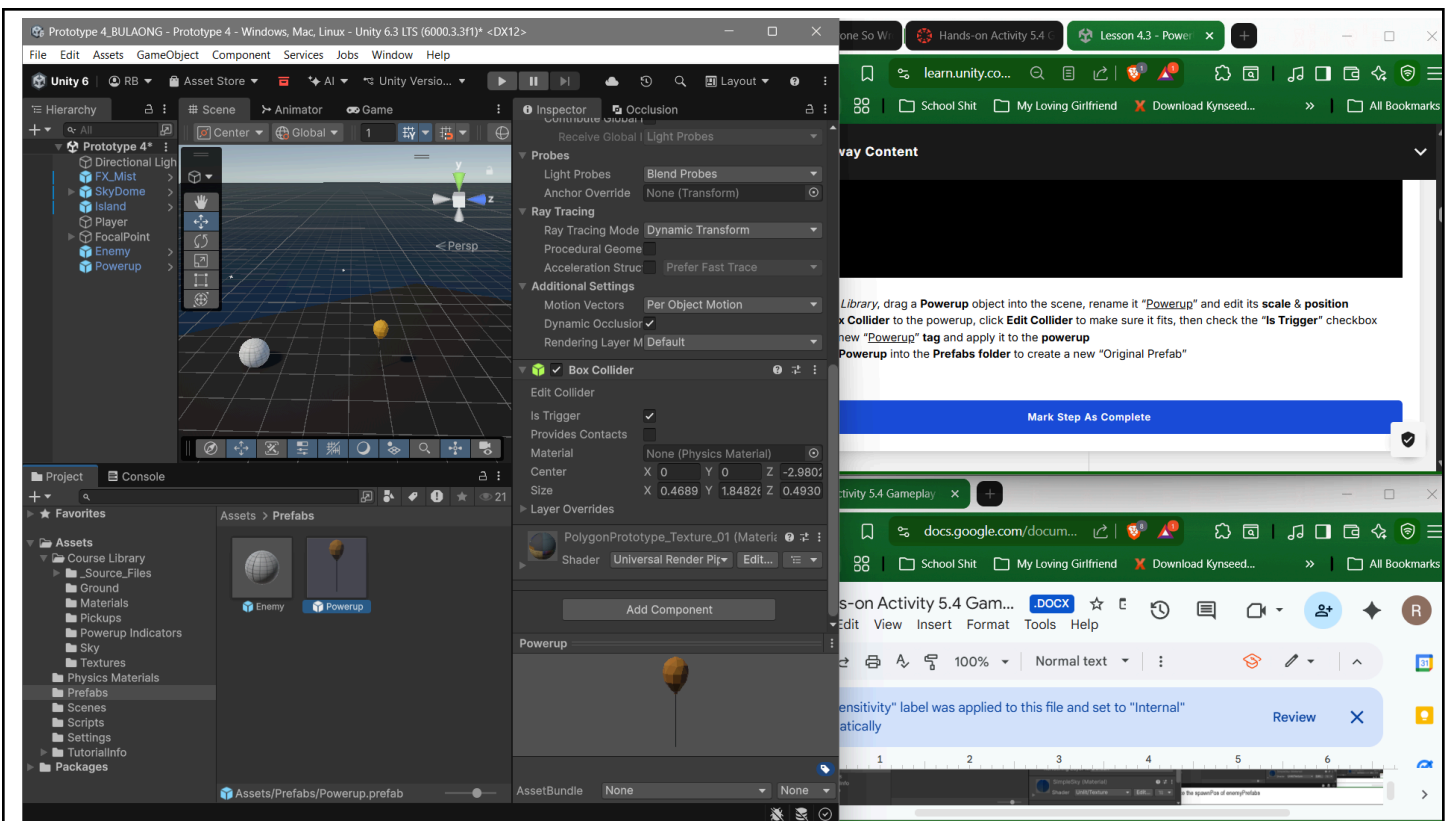
Create a spawn manager script



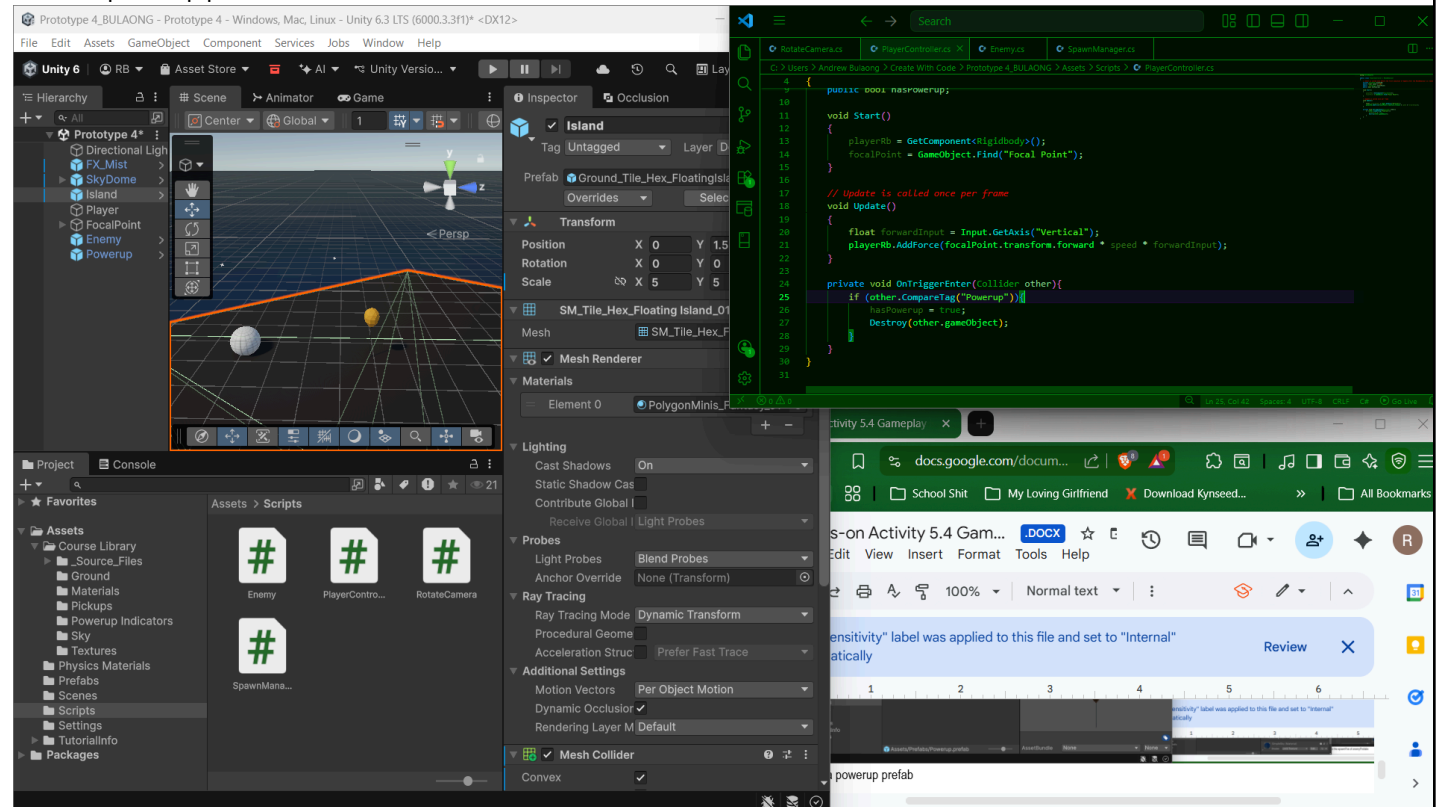
Randomize the spawnPos of enemyPrefabs



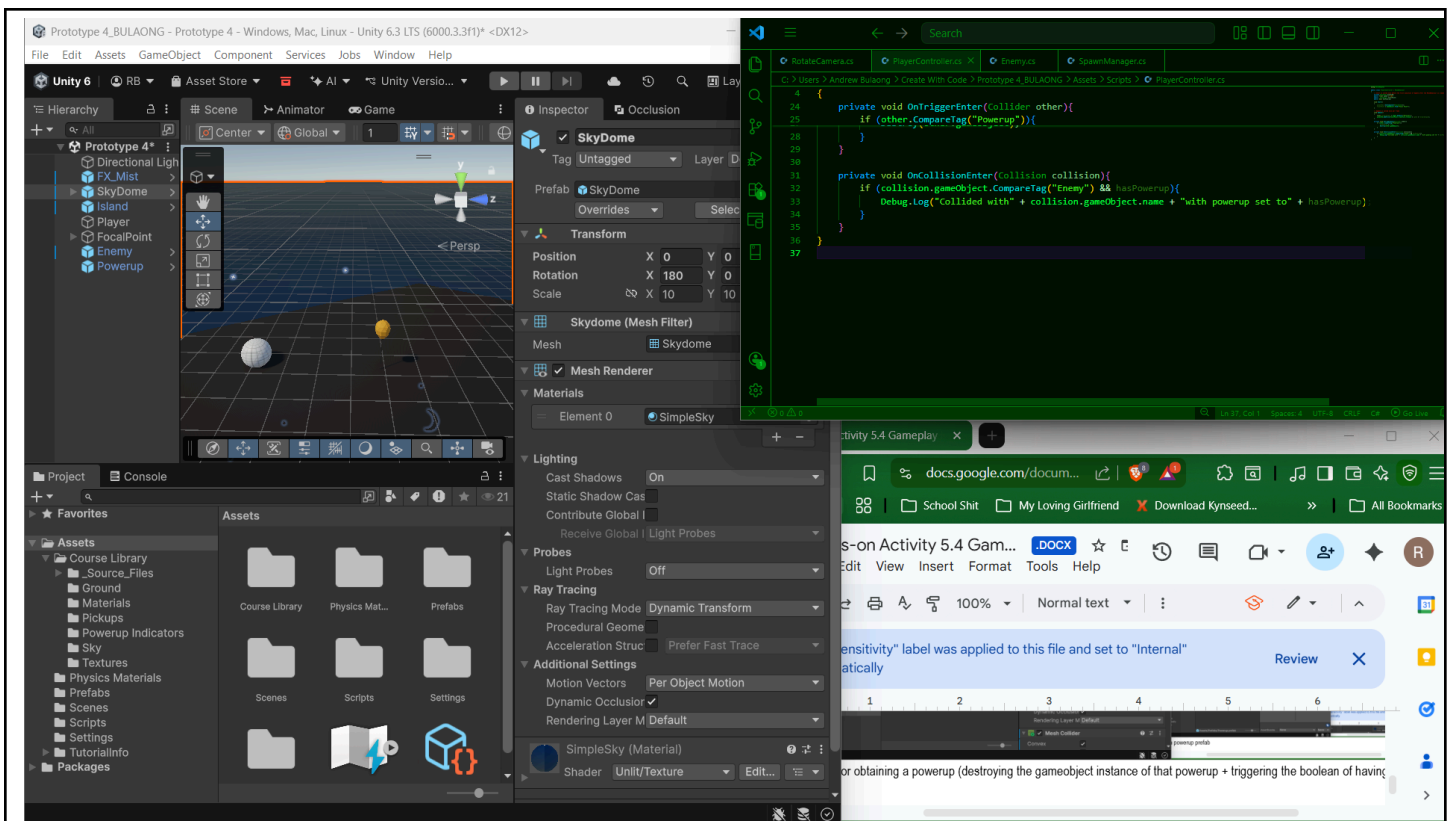
Create a function for generating random spawn positions instead of attaching it directly to start function



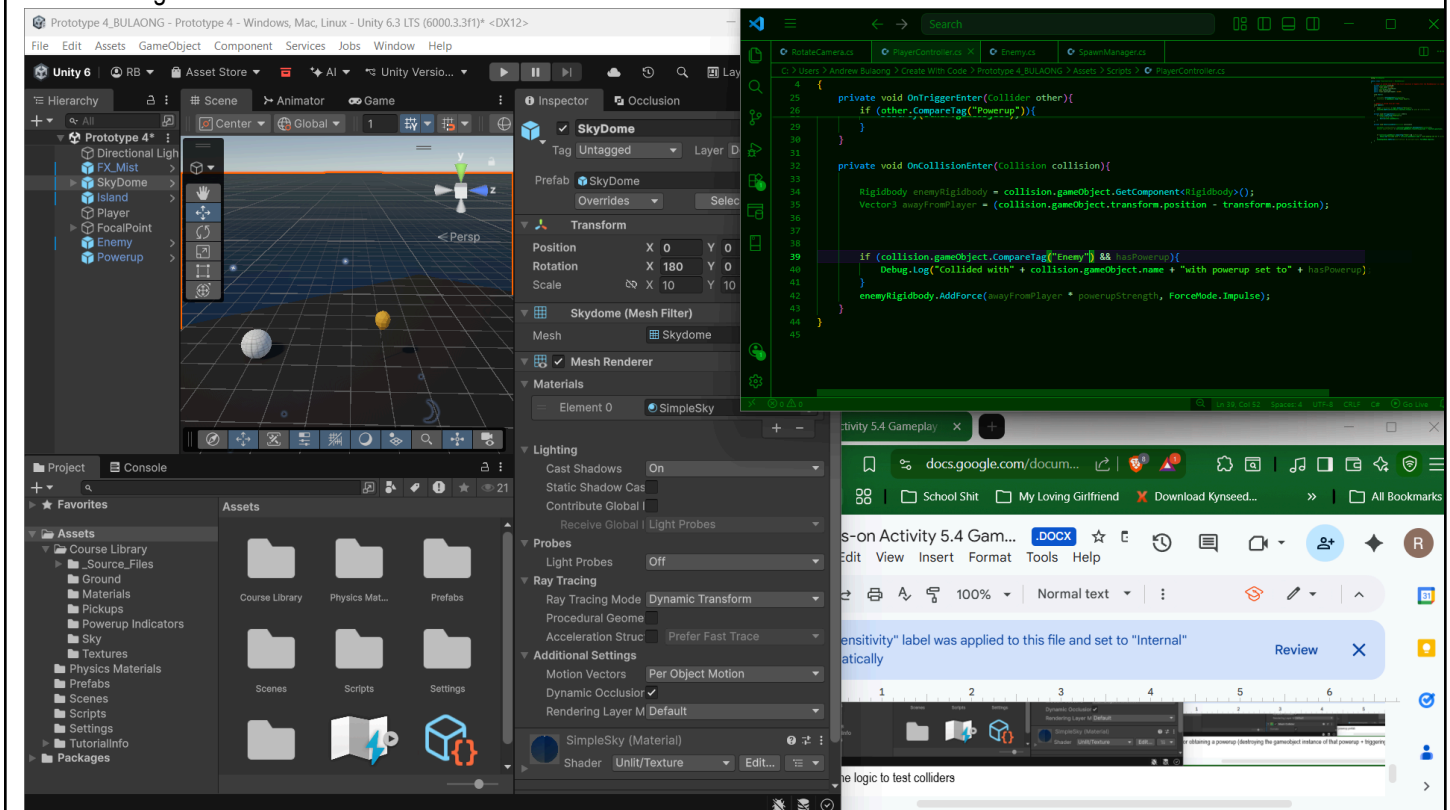
Create a powerup prefab



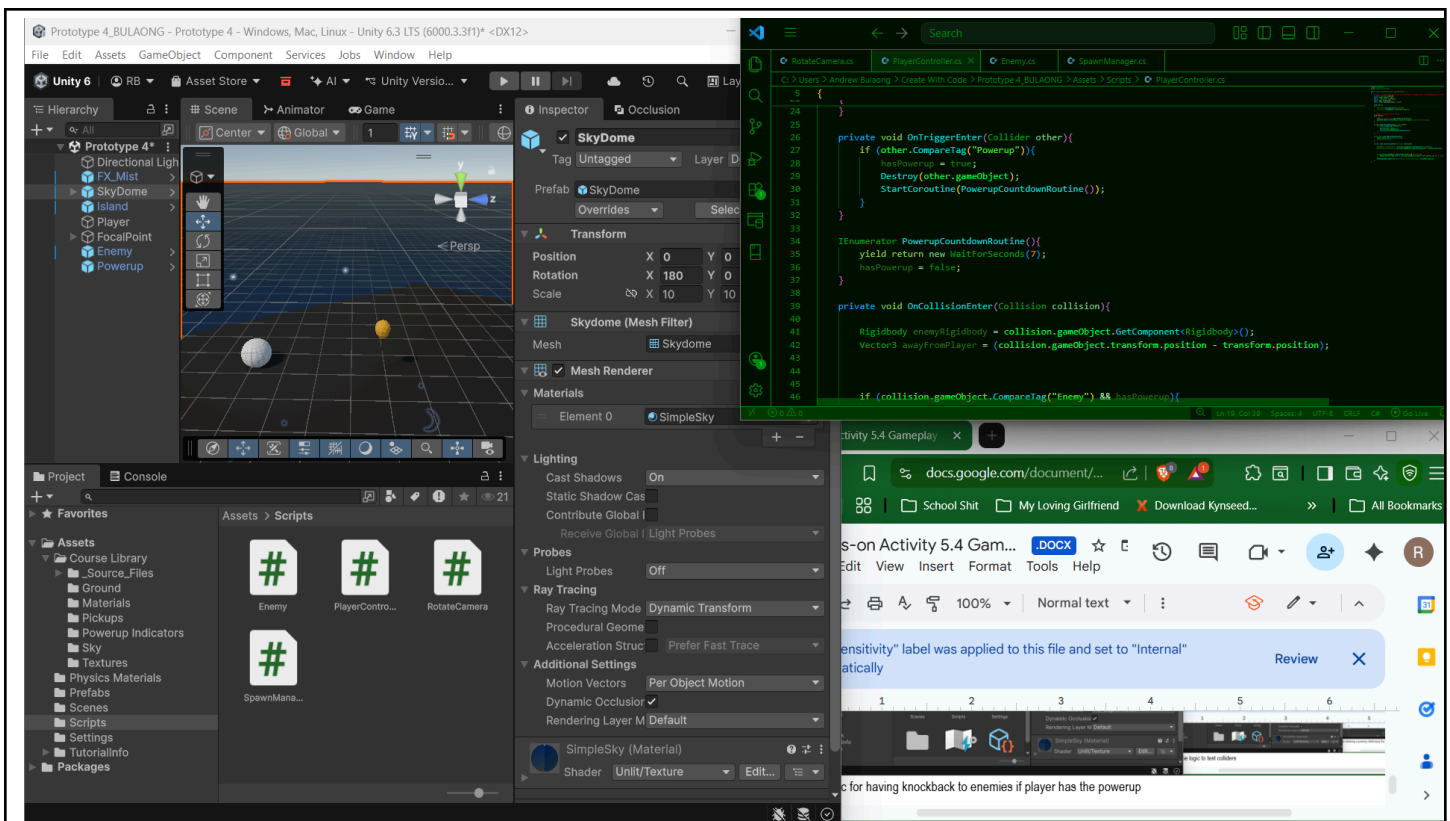
Add logic for obtaining a powerup (destroying the gameobject instance of that powerup + triggering the boolean of having a powerup as true)



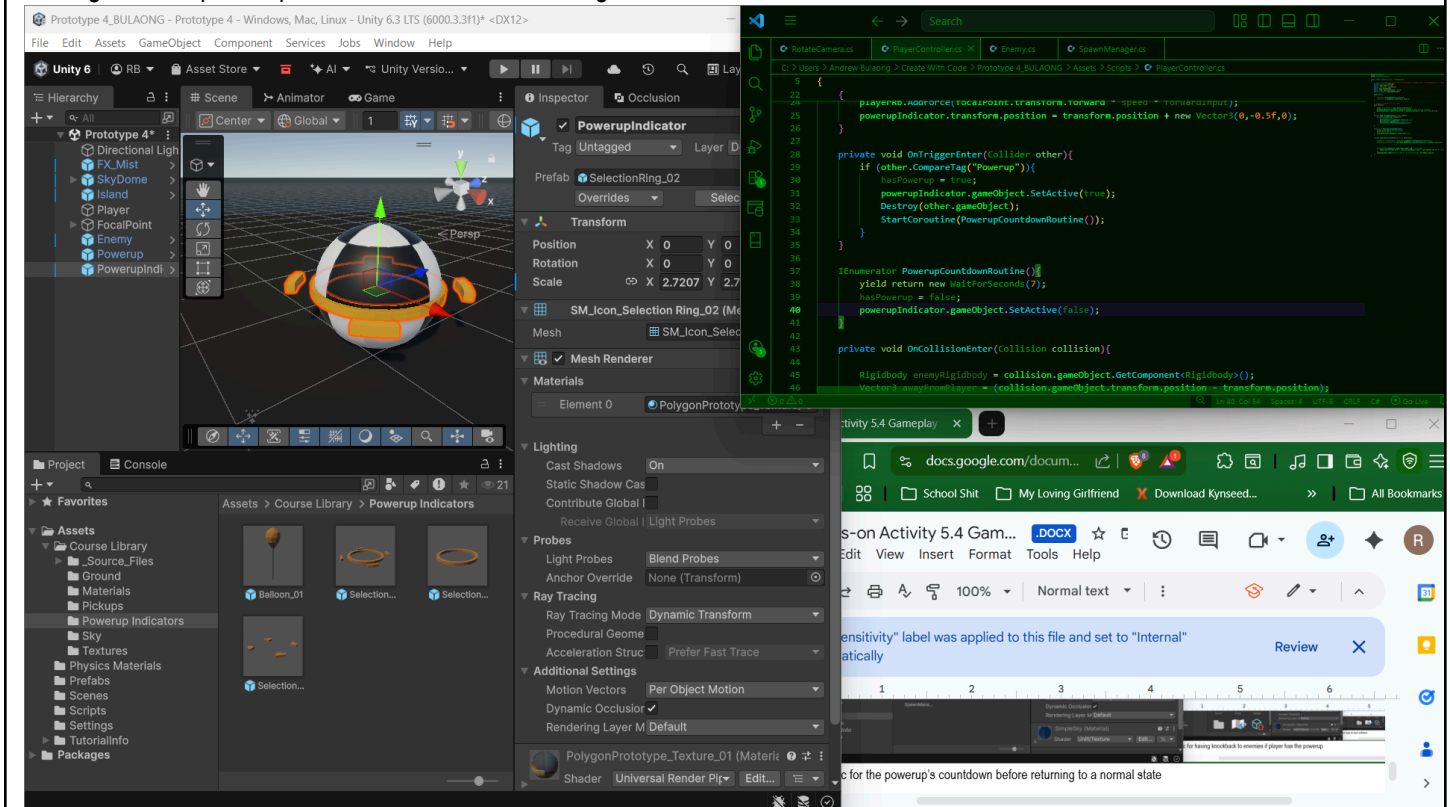
Add some logic to test colliders



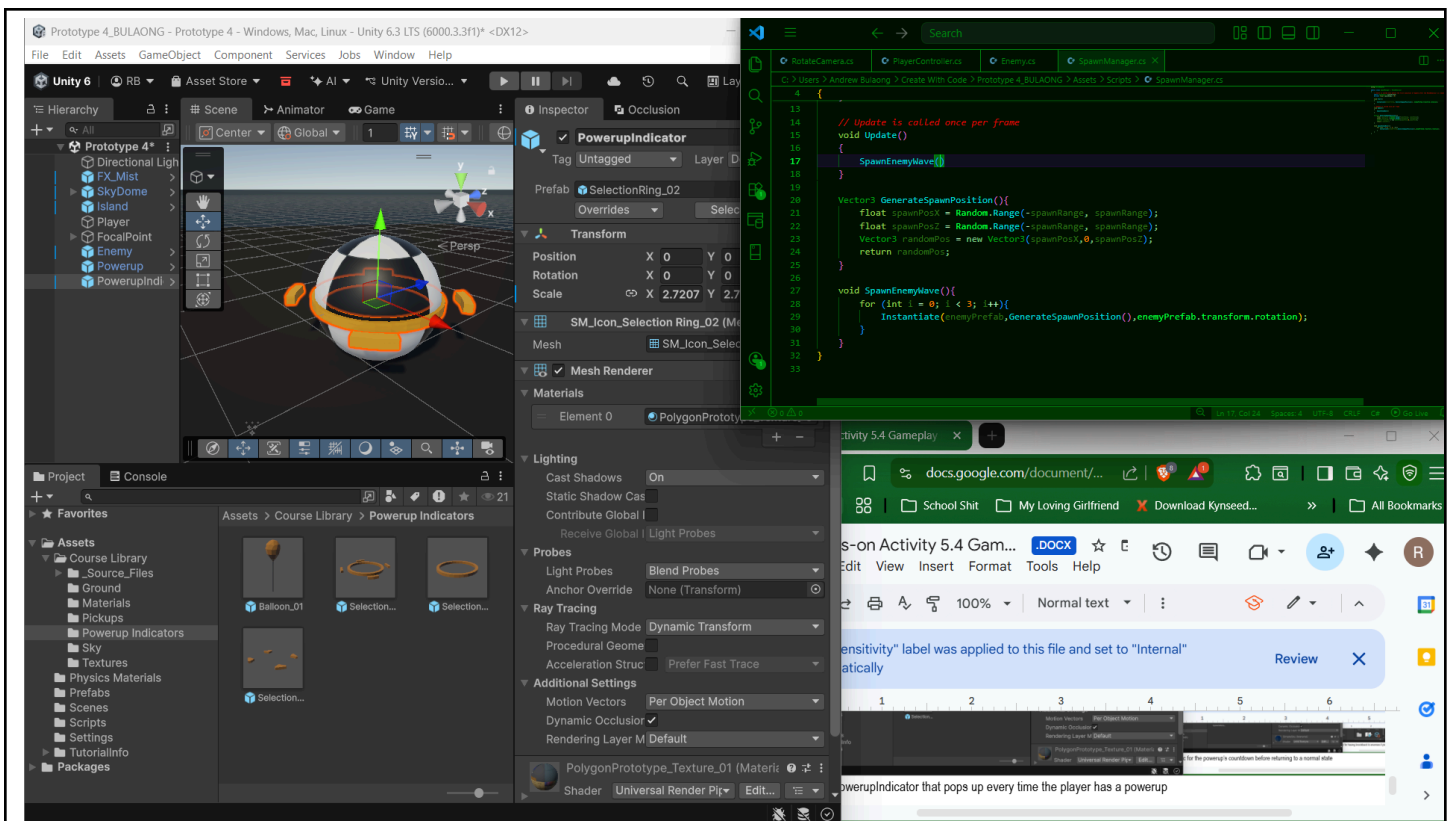
Add logic for having knockback to enemies if player has the powerup



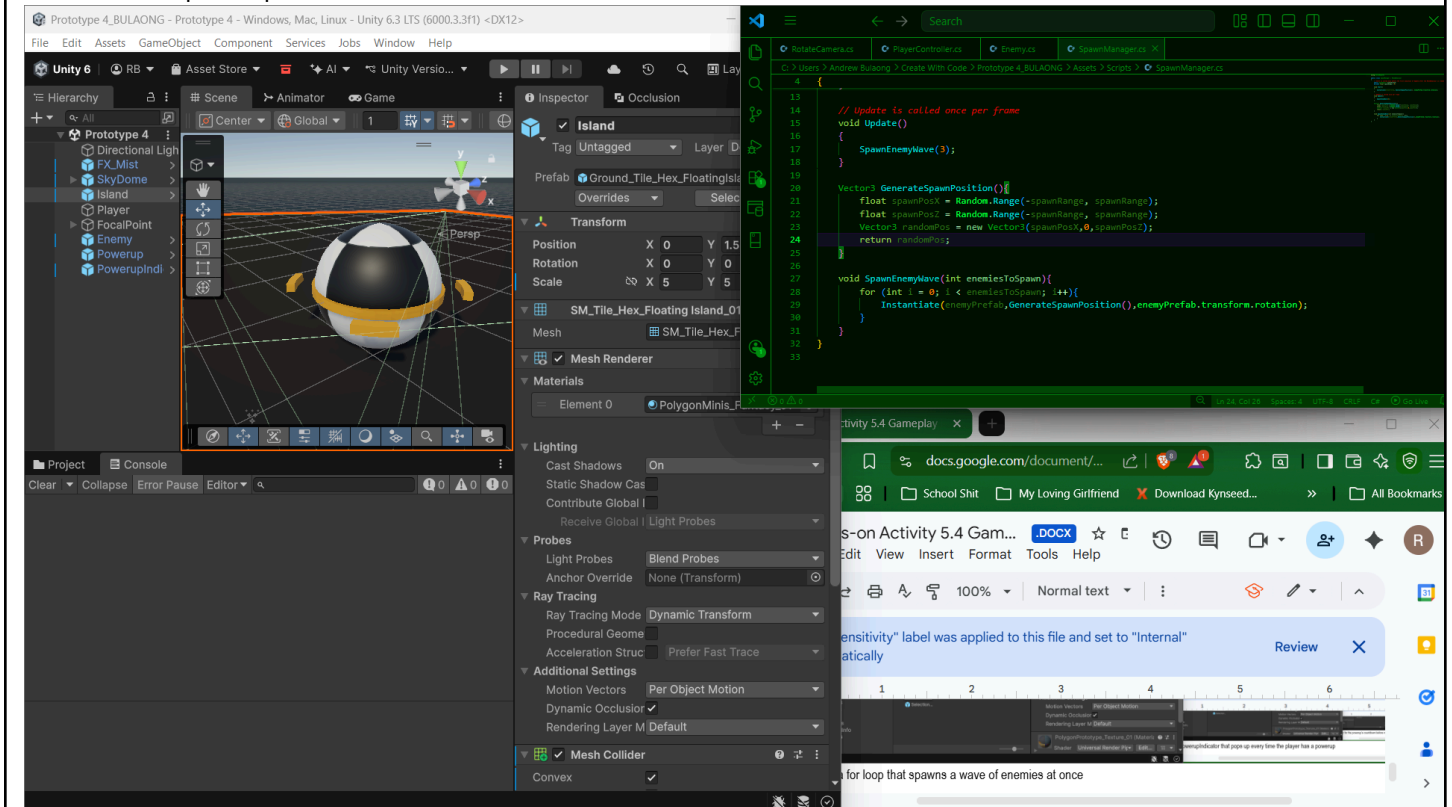
Add logic for the powerup's countdown before returning to a normal state



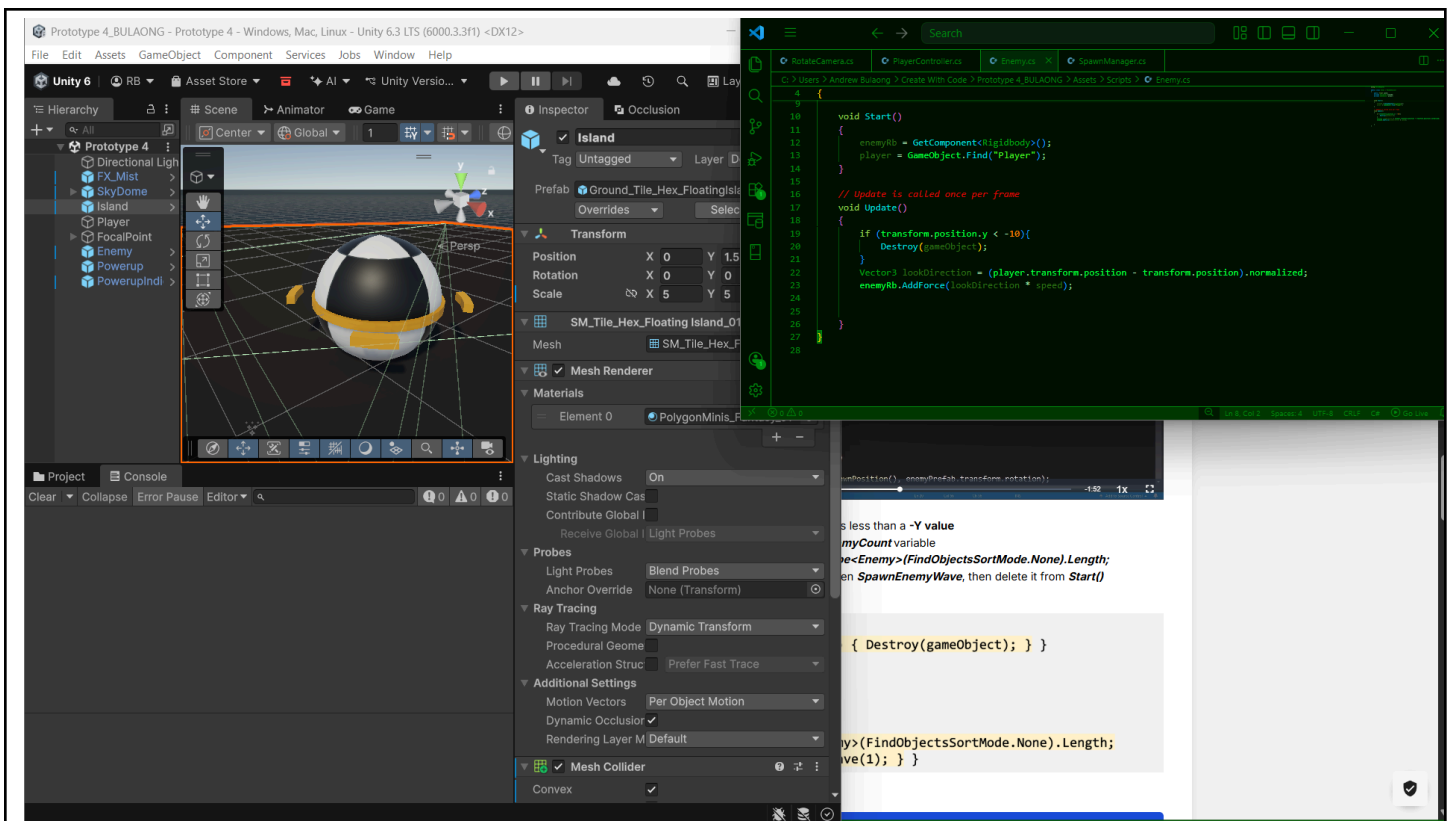
Add a powerupIndicator that pops up every time the player has a powerup



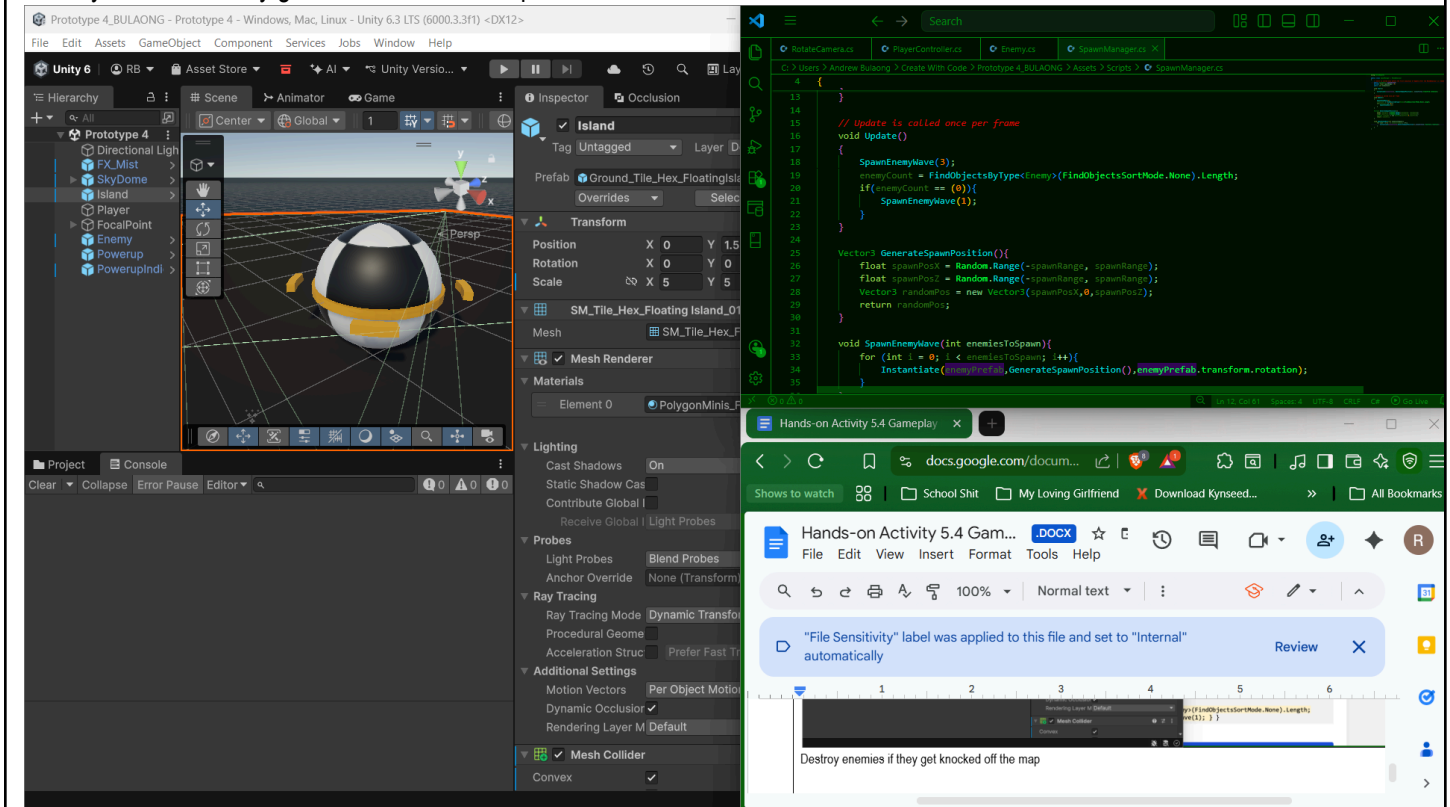
Create a for loop that spawns a wave of enemies at once



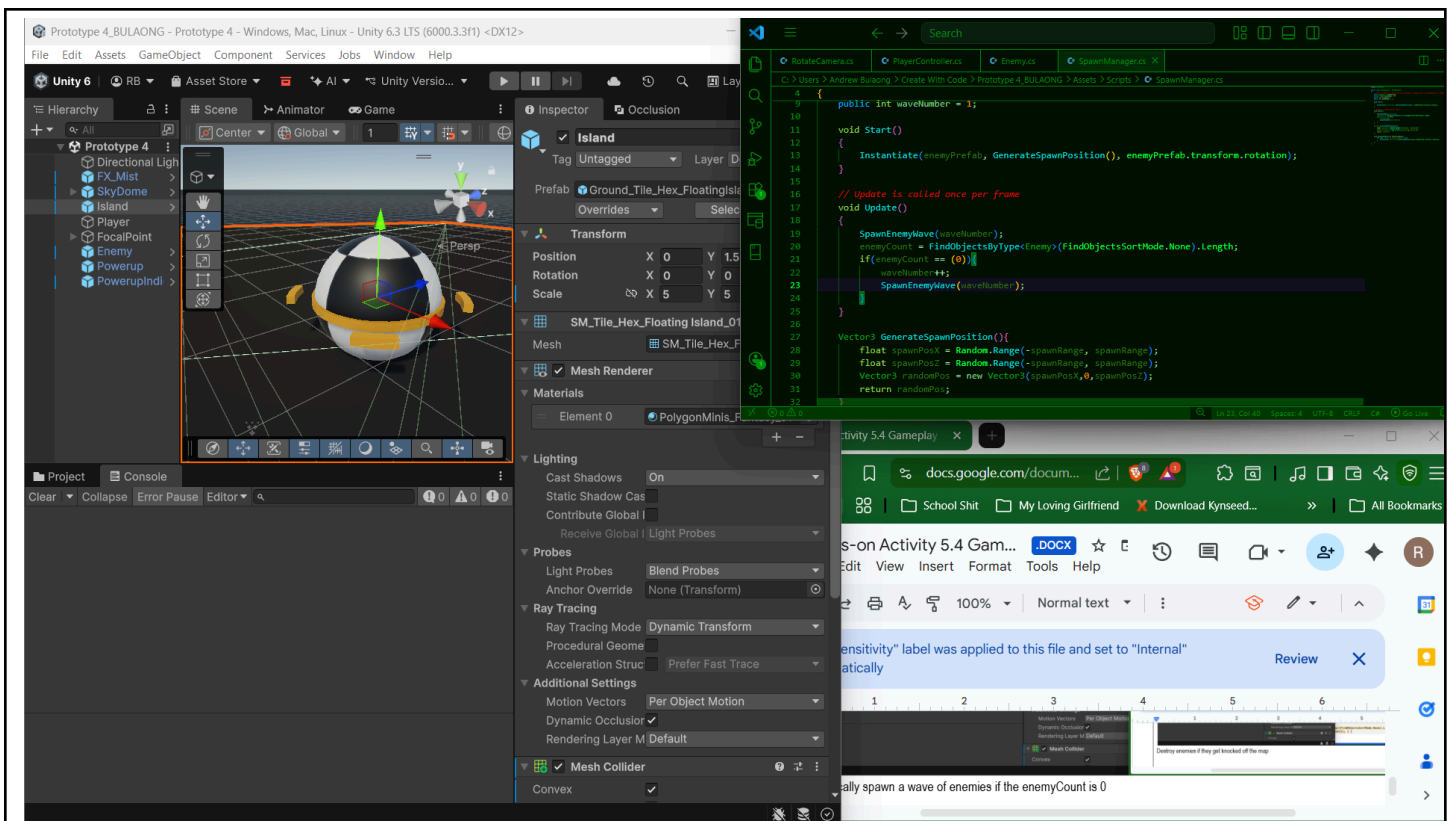
Add a parameter that freely lets us toggle how many enemies to spawn



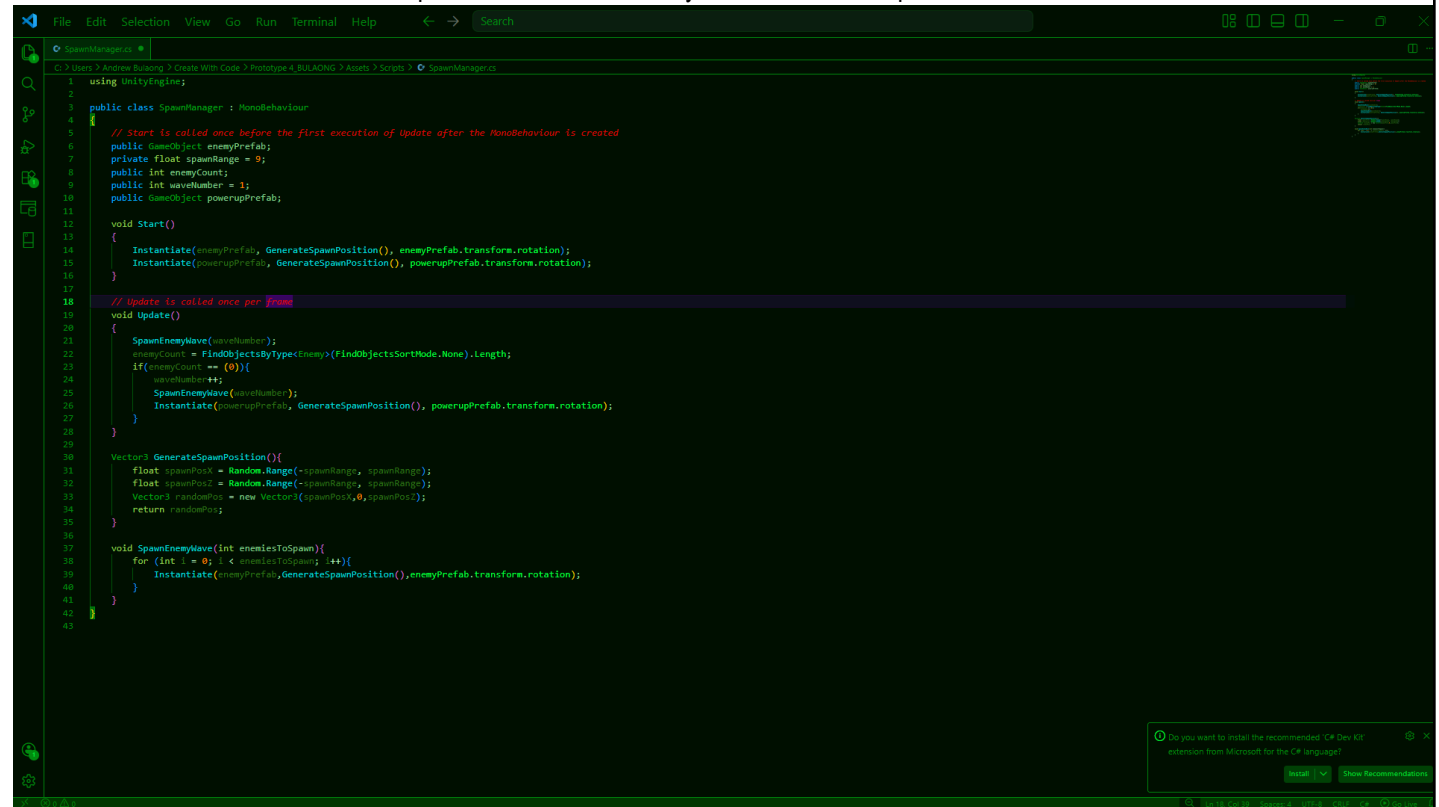
Destroy enemies if they get knocked off the map



Automatically spawn a wave of enemies if the enemyCount is 0



Increase the amount of enemies that spawn based on how many waves had been spawned beforehand



generate powerups once per wave

Your results

Congratulations! You passed. If you would like to score higher, you can [retake](#) the quiz. We always save your highest score.

Passed • You got 9 of 10 correct



+60 XP

You gained 60 XP points towards
Programming

You completed this quiz on April 17, 2026

Next up in Junior Programmer



User Interface

+850 XP • 7h 30m

[Continue to next](#)

Quiz

4. Conclusion

After completing the activity, I understood what the IEnumerator function does, as well as other new knowledge I can use to make my own scripts such as Coroutines and the technique of using randomly generated numbers to spawn objects at random. Lastly, as I progress more and more through the different activities that Unity Learn's Junior Programmer course provides, I can clearly see the "best practices" that professional game developers use in games.

5. Assessment Rubric

Lab Activity Rubric



Criteria	Ratings						Pts
SO 7 PI 1 Student Outcome 7.1 Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently	4 pts Satisfactory Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor Relies on classroom instruction only	1 pts Very Poor No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 2 Student Outcome 7.2 Learn independently threshold: 4.8 pts	6 pts Excellent Completes an assigned task independently and practices continuous improvement	5 pts Good Completes an assigned task without supervision or guidance	4 pts Satisfactory Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory Requires detailed or step-by-step instructions to complete a task	2 pts Poor Shows little interest to complete a task independently	1 pts Very Poor No interest to complete a task independently	6 pts
SO 7 PI 3 Student Outcome 7.3 Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory Apply the gathered information to formulate the problem	2 pts Poor Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor Gather information from a variety of sources	6 pts
SO 7 PI 4 Student Outcome 7.4 Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory Shows some creative ways to solve the problem	2 pts Poor Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor Ideas are copied or restated from the sources consulted	6 pts

Total Points: 24